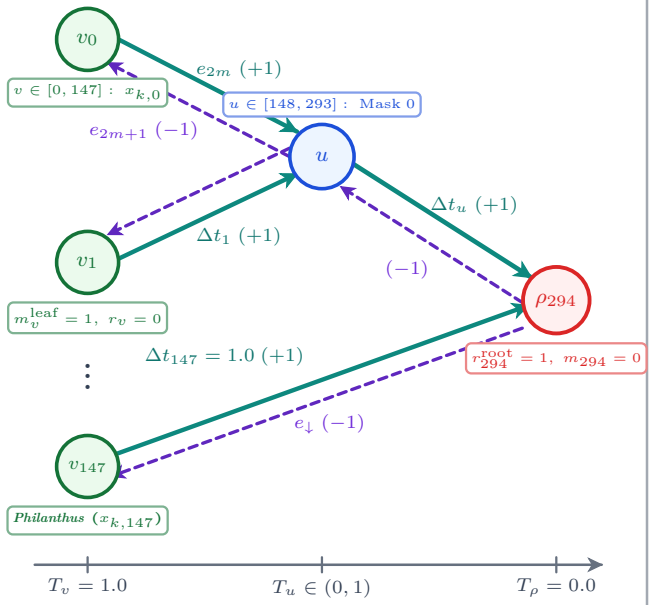
(A) $\mathcal{T}_{\text{Newick}} \mapsto \mathcal{G}_{\mathcal{T}} = (V, E_{\text{dir}}),\quad |V| = 295,\ |E_{\text{dir}}| = 588$



$V_{\text{leaf}} = \{0, \ldots, 147\},\ V_{\text{int}} = \{148, \ldots, 294\}$ (Postorder)
$e_{\uparrow} = (v \to u)\ (\text{dir}_e = +1)\ \oplus\ e_{\downarrow} = (u \to v)\ (\text{dir}_e = -1)$

(B) Explicit PyG Tensors: $\mathbf{X}_k$, edge_index, edge_attr

1. Node Tensor $\mathbf{X}_k \in \mathbb{R}^{295\times 6}$ ($\delta_{k,v} = x_{k,v}^{\text{obs}} - \bar{x}_k,\ \tilde{d}_v = d_{\text{topo}}(v)/16$):

$v \in V$	$x_{k,v}^{\text{obs}}$	$\delta_{k,v}$	m_v^{leaf}	r_v^{root}	T_v	$\tilde{d}_v$
0 . . . 147 (Leaf)	$x_{k,v}$	$x_{k,v} - \bar{x}_k$	1	0	1.000	$d_v/16$
148 . . . 293 (Int)	0.0 (Mask)	0.0 (Mask)	0	0	T_u	$d_u/16$
294 (Root ρ)	0.0 (Mask)	0.0 (Mask)	0	1	0.000	0.000

2. COO Connectivity edge_index $\in \mathbb{Z}^{2\times 588}$ ($|E_{\text{dir}}| = 2 \times 294 = 588$):

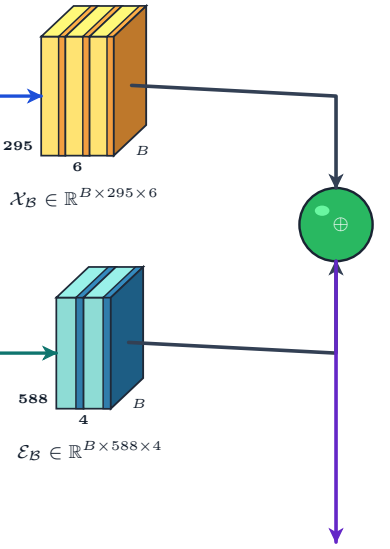
Row	e_0 (↑)	e_1 (↓)	⋯	e_{2m} (↑)	e_{2m+1} (↓)	e_{587} (↓)
src (0)	v_0 (0)	u_{148}	⋯	child v	parent u	ρ (294)
dst (1)	u_{148}	v_0 (0)	⋯	parent u	child v	v_{147} (147)

3. Branch Tensor edge_attr $\in \mathbb{R}^{588\times 4}$ ($\mathbf{e}_{uv} \in \mathbb{R}^4$):

$e \in E_{\text{dir}}$	Δt_e	$(\Delta t_e + 10^{-3})^{-1}$	$\ln(\Delta t_e)$	dir_e
e_{2m} $(v \to u)$	Δt_v	$(\Delta t_v + 10^{-3})^{-1}$	$\ln \Delta t_v$	+1.0 (↑)
e_{2m+1} $(u \to v)$	Δt_v	$(\Delta t_v + 10^{-3})^{-1}$	$\ln \Delta t_v$	−1.0 (↓)

c_{cond} : Stage I-A (σ_k), Stage I-B (μ_k), Stage II Joint ($c_{\text{cond}} \equiv 0.0$, 147 DOF)

(C) 3D Batched PyG Tensors ($\mathcal{B}$)



PyG Mini-Batch $\mathcal{B}$ (B Trees):
Data($\mathbf{X}_{\mathcal{B}}$, edge_index, edge_attr)
$\mathbf{X}_{\mathcal{B}} \in \mathbb{R}^{(295B)\times 6},\quad \mathbf{E}_{\mathcal{B}} \in \mathbb{R}^{(588B)\times 4}$
$\mathcal{M}_1 \to \mathcal{M}_{5B}$: $\mathcal{G}_{\mathcal{T}} \mapsto (\hat{\mu}_k, \hat{\sigma}_k)_{k=1}^{B}$